

Lec10

Bootstrap

Bootstrap is a popular **front-end CSS framework** used to design responsive and mobile-first websites quickly.

It provides ready-made:

- CSS styles
- Layout system (Grid)
- UI components
- JavaScript plugins

So instead of writing all CSS from scratch, you can use Bootstrap's predefined classes.

Why use Bootstrap?

Faster web development

Responsive design (works on mobile, tablet, desktop)

Consistent UI across browsers

Pre-built components like buttons, forms, cards, navbars, modals, etc.

Easy to learn and use

How to add Bootstrap?

```
<link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css"  
rel="stylesheet">
```

<https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css/bootstrap.min.css>

Advantages of Bootstrap

Easy to use

Mobile-first design

Large community support

Saves development time

Cross-browser compatibility

Pre-designed UI components

Disadvantages of Bootstrap

Websites may look similar if heavily relying on default styles.

Extra CSS/JS files can increase page size.

Customization may require additional CSS.

Task

Build a Bootstrap-based responsive webpage and optimize Bootstrap code and improve layout quality